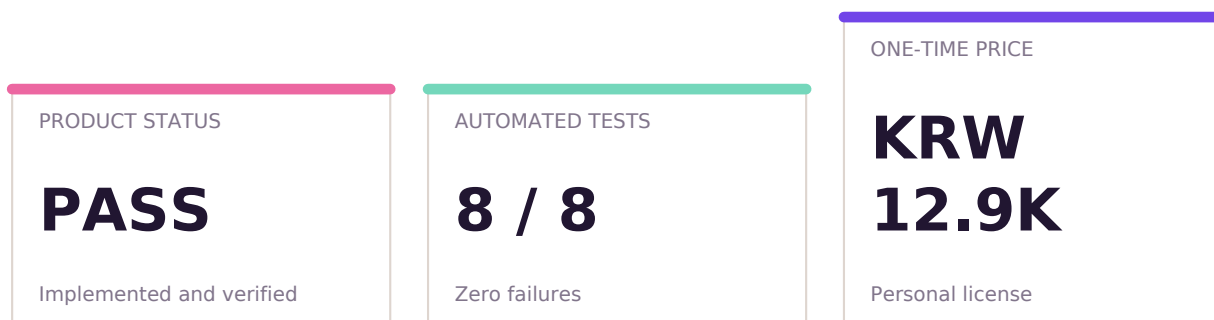


SYSTEM FOUNDRY / PERSONAL MVP

FilamentFair

A local-first print ledger that prices material, energy, machine time, failure reserve, and margin.



01 / PROJECT OVERVIEW

Project overview

FilamentFair turns slicer estimates into a practical true-cost ledger. A maker enters filament use, print time, power, machine-time cost, failure reserve, and margin to see a cost breakdown and a rounded suggested price.

02 / TARGET CUSTOMER

Target customer

Hobbyist 3D printer owners who make replacement parts, props, prototypes, gifts, or occasional paid requests for friends and local communities.

03 / PROBLEM

Problem

Charging only for filament ignores electricity, printer wear, long machine occupation, failed attempts, and a sustainable margin. Makers either lose money or choose an arbitrary price they cannot explain.

PRODUCT

Three focused jobs

04 / CORE FEATURES

Core features

The MVP implements one complete print-costing loop:

01	Validated print job intake Capture material, grams, hours, spool price, power, electricity rate, machine cost, reserve, margin, and status.
02	True-cost breakdown Separate material, energy, machine time, and failure reserve instead of hiding them in one guess.
03	Suggested price and job ledger Apply target margin, round to a usable price, and separate planned, completed, and failed work.

05 / HOW TO USE

How to use

Copy grams and print hours from the slicer, set the current spool price and printer assumptions, then choose a failure reserve and target margin. Review the breakdown before accepting a request and mark the job completed or failed afterward.

06 / SYSTEM ARCHITECTURE

System architecture

A static browser UI calls pure validation and cost functions and stores the job ledger only in localStorage. Node tests exercise the same module. No account, backend, database, analytics, payment system, or external API is used.

Browser UI	Cost module	Local storage	Node tests
HTML + CSS	Rules + pricing	Device-only data	Pure functions

07 / TECHNOLOGY

Technology

Static HTML, responsive CSS, vanilla JavaScript, localStorage, bundled Noto Sans KR fonts, and the built-in Node.js test runner. Runtime dependencies: zero. Paid APIs: zero.

VALIDATED INTERFACE

Actual running screen

Captured from headless Chromium after a cosplay clip was validated and priced.

FilamentFair
PRICE EVERY PRINT FAIRLY.

생물 복원

출력 작업을 등록하고 비용 원장을 업데이트했습니다.

MAKER COST DESK

메이커의 시간까지, 가격에 넣자.

재료비만 받고 손해 보는 3D 출력을 막아보세요.

견적 중 작업

3

PLANNED

총 사용량

535g

FILAMENT

예상 원가

₩28,903

TRUE COST

권장 가격 합계

₩43,100

반포 견적 ₩3,600

PRINT LEDGER

출력 비용 원장

상태와 원가를 한눈에

견적 중 PETG - 260g	Desk Organizer 12시간 · 실재 예비 10% · 목표 마진 30% 재료 ₩6,240 전기 ₩288 장비 ₩5,000	예상 원가 ₩13,781	권장 가격 ₩19,700	견적 중 저장
견적 중 PLA - 380g	Miniature Tower 8.5시간 · 실재 예비 15% · 목표 마진 35% 재료 ₩4,320 전기 ₩204 장비 ₩4,250	예상 원가 ₩10,090	권장 가격 ₩15,600	견적 중 저장
견적 중 PLA - 95g	Cosplay Clip 4시간 · 실재 예비 15% · 목표 마진 35% 재료 ₩2,280 전기 ₩96 장비 ₩2,000	예상 원가 ₩5,032	권장 가격 ₩7,800	견적 중 저장
출력 완료 ABS - 45g	Replacement Knob 2.2시간 · 실재 예비 20% · 목표 마진 25% 재료 ₩1,080 전기 ₩53 장비 ₩1,100	예상 원가 ₩2,679	권장 가격 ₩3,600	출력 완료 저장

NEW PRINT

작업 등록

기본값을 내 프린터와 필라멘트에 맞춰 조정하세요.

작업명

예: Cosplay Clip

소재

PLA

사용량(g)

95

출력시간(h)

4

1kg 가격

24000

소비전력(W)

120

전기요금/kWh

200

장비시간 비용/h

500

실재 예비율(%)

15

목표 마진(%)

35

상태

견적 중

비용 계산 · 등록

LOCAL FIRST

작업 데이터는 이 브라우저에만 저장됩니다.

08 / RUNNING SCREEN

Running screen

The verified screen shows three planned jobs using 535 g, KRW 28,903 estimated cost, and KRW 43,100 suggested price after adding a 95 g cosplay clip. Marking it printed returned planned totals to baseline and increased completed estimated revenue to KRW 11,400.

VERIFICATION

Evidence, not assumptions

09 / TEST RESULTS

Test results

Every required check was executed against the generated project.

Check	Result	Evidence
Dependency installation	PASS	npm install completed; zero runtime dependencies
Application start	PASS	README HTTP server returned status 200
Core workflow	PASS	Planned price KRW 35.3K to 43.1K to 35.3K
Automated tests	8 / 8	Validation, breakdown, reserve, price, status
README commands	PASS	Install, test, and start commands executed
Credential scan	PASS	No secret patterns and no .env file
PDF	PASS	Generated, parsed, rendered, visually inspected

10 / SECURITY AND LIMITATIONS

Security and limitations

FilamentFair sends no data outside the browser and requires no credentials. Text and numeric inputs are validated, and displayed text is HTML-escaped. Device-local data is unencrypted and disappears when browser storage is cleared. Results depend on user assumptions and exclude tax, delivery, post-processing, design labor, and local business obligations. Users must follow the printer and material manufacturer's safety guidance.

GO TO MARKET

Stop giving machine time away

11 / PRICING AND REVENUE MODEL

Pricing and revenue model

Sell a downloadable personal license for KRW 12,900 as a one-time purchase. The price can be recovered by correctly pricing one medium print request. The zero-hosting design keeps operating cost low and supports a simple downloadable product.

12 / FIRST CUSTOMER ACQUISITION

First customer acquisition

Recruit twenty hobbyists from Korean 3D printing, model-making, cosplay, and maker communities. Ask them to price three recent jobs with and without machine time and failure reserve. Convert with the visible difference between filament-only and true cost.

13 / IMPROVEMENT PLAN

Improvement plan

Priorities after observing real costing habits:

- Reusable printer and filament presets.
- JSON backup and CSV quote export.
- Design labor, post-processing, packaging, and delivery options.
- Optional currency and tax profiles without payment processing.

PRODUCT BOUNDARY

FilamentFair should remain a transparent estimating tool. It should not claim to provide tax advice, guarantee profit, control a printer, or replace safe equipment operation and supervision.

ASSESSMENT

Overall evaluation

14 / OVERALL SCORE

Overall score

Six dimensions are scored out of 10. A higher maintenance score means easier maintenance.

Dimension	Score	Rationale
Customer problem strength	8/10	Underpricing long prints creates immediate, measurable loss.
One-day completeness	9/10	The intake, breakdown, pricing, and status loop is tested.
Sales potential	8/10	One correctly priced request can justify the license.
Maintenance difficulty	9/10	Zero runtime dependencies and no backend keep maintenance light.
Differentiation	8/10	Machine time and failure reserve improve a filament-only calculator.
Scalability	8/10	Presets, exports, labor, delivery, and currencies can expand it.

OVERALL

8.3 / 10**Proceed to a twenty-maker pricing trial.**

Verdict: technically complete and visually polished for a personal MVP. The next useful evidence is whether users change their quoted prices after seeing machine time and failed-print exposure.